

Enhancement Final Report

Phase 1 NS-3 simulation enhancement — completion report

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- **Lab:** PIC Lab, KIT
- **Generated:** 2026-09-27 22:50
- **Plan:** NS-3 Simulation Enhancement Plan (four-item accuracy upgrade)
- **September campaign:** Traffic_qos_matrix_sep_seeds7to16 — **640** seeded scenarios

1. Purpose

This report closes the four enhancement items that bridge Phase 1 hybrid simulation results toward field-validation readiness: realistic mobility, robot-oriented traffic QoS, WiFi-mesh internal handover / Guard Timer, and statistical reinforcement of the WiFi+LTE vs WiFi+5G comparison.

2. Status of the four enhancement items

#	Item	Target	Primary deliverables	Status
1	Realistic mobility	Jun 2026	waypoint_mobility scenarios, Waypoint_outputs/mobility_comparison_report.pdf	Complete
2	Traffic model (Control/Sensor/Video)	Jul 2026	traffic_qos.cc, flow_metrics.py, July traffic QoS report / presentations	Complete
3	WiFi mesh internal HO + Guard Timer	Aug 2026	Intra-mesh logging, guard_timer_report.pdf	Complete
4	Statistical reinforcement (10 seeds)	Sep 2026	640-run matrix, stats_analysis.py, this report	Complete

2.1 Mobility (Item 1)

Gauss-Markov random walks were supplemented with waypoint + dwell patterns that better match patrol / transport / work robots on a construction site. Comparison outputs (trajectory plots, RSSI heatmaps, switching timelines, and the mobility comparison PDF) live under Waypoint_outputs/.

2.2 Traffic QoS (Item 2)

Generic HTTP/video mixes were replaced by three DSCP-marked robot flows (Control / Sensor / Video) with per-flow FlowMonitor metrics. The September matrix reuses that traffic model across **640** cells (2 modes × 2 bands × 4 STA × 4 payloads × 10 seeds).

2.3 WiFi internal handover / Guard Timer (Item 3)

Intra-mesh HO events are classified in the switch log (type field: intra_mesh / wifi_to_cell / cell_to_wifi). Guard Timer experiments and the review PDF are under Traffic_qos_outputs/guard_timer_study/ and august_deliverable/.

2.4 Statistical reinforcement (Item 4) — headline results

After expanding from 3 to **10 RNG seeds**, Bootstrap CIs, Mann-Whitney U, and Kruskal-Wallis tests were applied to the September campaign.

Control KPI	LTE mean [95% CI]	5G NR mean [95% CI]	MW p (overall)	Significant?
PDR (%)	85.640 [84.702, 86.580]	67.951 [66.497, 69.435]	1.09614e-59	yes
Mean delay (ms)	42.919 [40.505, 45.275]	17.345 [16.347, 18.346]	6.20064e-58	yes

- Mann-Whitney tests with $p < 0.05$: **46 / 60**

- Kruskal-Wallis tests with $p < 0.05$ (STA trend): **33 / 36**

Detailed tables and figures are in Sections 4–6 below (same content as `stats_analysis.md`).

3. Cross-item conclusions

1. **Mobility realism** reduces the mismatch between simulated RSSI traces and path-constrained robots before field trials.
2. **Per-flow QoS** makes Control latency/loss requirements directly measurable instead of hiding them in a blended PDR.
3. **Guard Timer / intra-mesh classification** separates mesh-internal roaming from cellular failover so the hybrid controller is not blamed for AP handovers.
4. **Ten-seed statistics** replace “appears different” language with Bootstrap intervals and non-parametric tests at $\alpha = 0.05$ for the LTE vs 5G NR claim.

Together these four items satisfy the September Weeks 3–4 deliverable set (`stats_analysis.py` + `Enhancement_Final_Report.pdf`) on top of the 640-run matrix completed in Weeks 1–2.

Appendix — Full statistical analysis

- **Author:** Sheikh Sayed Bin Rahman
- **Lab:** PIC Lab, KIT
- **Generated:** 2026-09-27 22:50
- **Campaign:** `/home/sayed/ns3_phase_2/ns-3.45/Traffic_qos_outputs/Traffic_qos_matrix_sep_seeds7to16`
- **Runs with metrics:** 640 (summary rows: 640, status ok/skipped: 640)
- **Bootstrap:** 1000 resamples, 95% percentile CI
- **Significance:** $p < 0.05$

1. Method

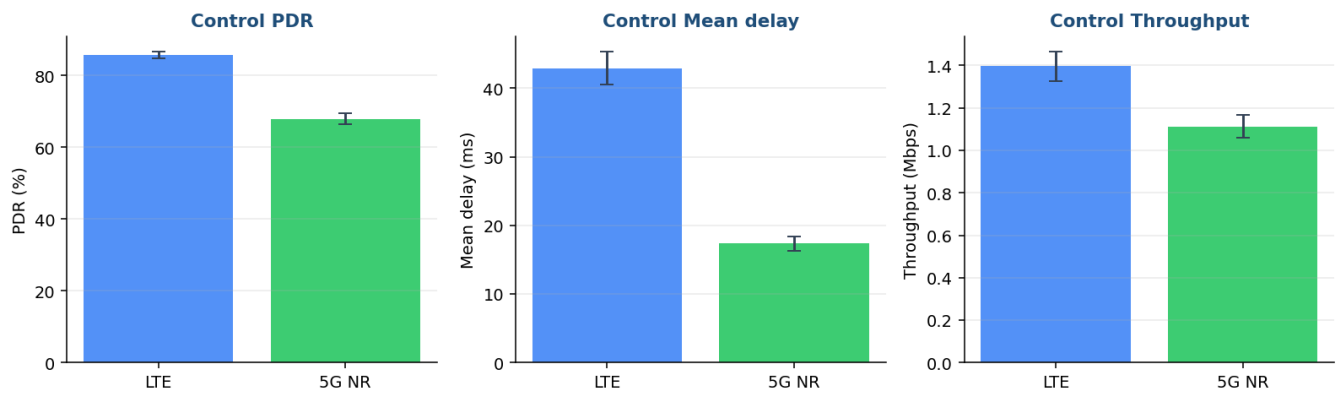
This script implements §5.2 of the NS-3 Simulation Enhancement Plan:

1. **Bootstrap 95% CIs** on the sample mean (1,000 resamples) for PDR, mean delay, P99 delay, and throughput, plus resolved switch interruption time.
2. **Mann-Whitney U** (two-sided) comparing WiFi+LTE vs WiFi+5G NR per flow (and per flow $\times$ STA).
3. **Kruskal-Wallis** across STA counts {5, 10, 15, 20}, with a Spearman ρ hint for monotonic direction.

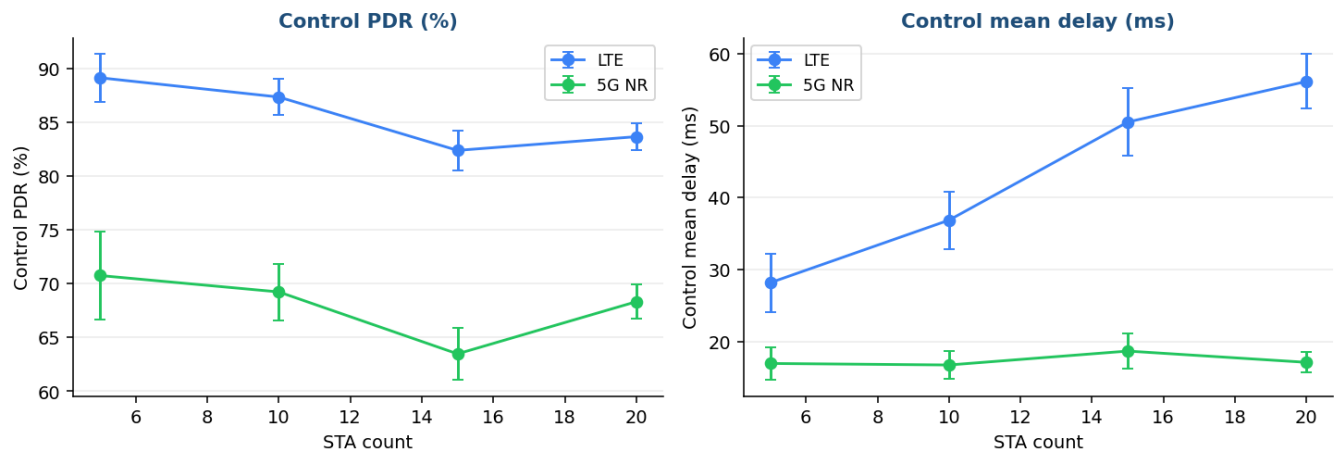
Unit of observation = one matrix cell (one seed $\times$ factor combination) per flow. With 10 seeds the LTE and NR arms each contribute 320 Control-flow observations in the overall mode comparison.

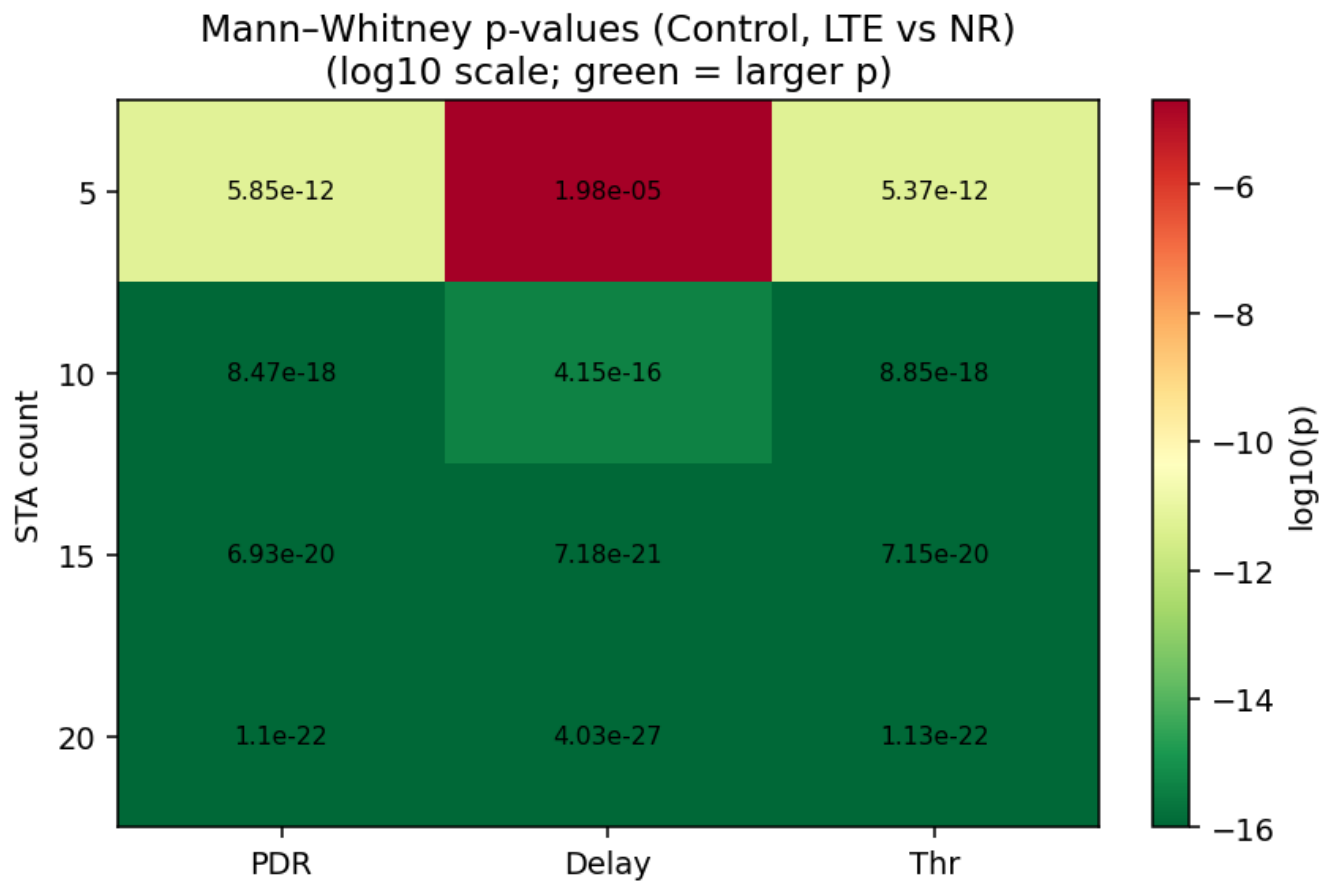
2. Figures

Control-flow KPIs — mean $\pm$ bootstrap 95% CI (10 seeds)



STA-count trends (mean $\pm$ approx. 95% SE) — Kruskal-Wallis input





3. Bootstrap 95% confidence intervals (Control flow)

Mode	KPI	n	Mean	95% CI	Half-width / mean	≤10% target
lte	PDR (%)	320	85.640	[84.702, 86.580]	0.011	yes
nr	PDR (%)	320	67.951	[66.497, 69.435]	0.022	yes
lte	Mean delay (ms)	320	42.919	[40.505, 45.275]	0.056	yes
nr	Mean delay (ms)	320	17.345	[16.347, 18.346]	0.058	yes
lte	Throughput (Mbps)	320	1.397	[1.328, 1.464]	0.049	yes
nr	Throughput (Mbps)	320	1.112	[1.058, 1.167]	0.049	yes
lte	P99 delay (ms)	320	452.516	[423.041, 480.627]	0.064	yes
nr	P99 delay (ms)	320	168.516	[156.876, 179.801]	0.068	yes

Switch interruption (resolved only)

Scope	Mode	STA	n	Mean (ms)	95% CI (ms)	% ≤ 200 ms
overall	ALL	ALL	24728	998.47	[958.30, 1041.15]	81.2
mode_lte	lte	ALL	12910	557.08	[527.40, 590.16]	85.5
mode_nr	nr	ALL	11818	1480.64	[1407.05, 1553.44]	76.5
sta_5	ALL	5	2024	1017.56	[855.50, 1189.42]	82.4
sta_10	ALL	10	5136	1096.05	[1007.46, 1184.05]	78.3
sta_15	ALL	15	7214	1062.27	[984.73, 1144.29]	80.6
sta_20	ALL	20	10354	901.88	[844.12, 954.21]	82.8

4. Mann-Whitney U — LTE vs 5G NR

$p < 0.05$ is marked **significant**. higher_median_mode is the arm with the better median (higher PDR/throughput, lower delay).

4.1 Overall by flow

Flow	KPI	n_LTE	n_NR	Median LTE	Median NR	U	p	Sig?	Better median
Control	PDR (%)	320	320	86.650	67.940	89307.5	1.09614e-59	yes	lte
Control	Mean delay (ms)	320	320	40.625	14.985	88726.0	6.20064e-58	yes	nr
Control	Throughput (Mbps)	320	320	1.327	1.063	65653.0	6.42934e-10	yes	lte
Control	P99 delay (ms)	320	320	446.500	139.500	84069.5	7.27451e-45	yes	nr
Sensor	PDR (%)	320	320	96.350	96.840	39684.5	8.49831e-07	yes	nr
Sensor	Mean delay (ms)	320	320	28.715	17.385	68605.0	9.93815e-14	yes	nr
Sensor	Throughput (Mbps)	320	320	7.277	7.389	50152.0	0.654237	no	nr
Sensor	P99 delay (ms)	320	320	164.000	124.000	58678.0	0.00138756	yes	nr
Video	PDR (%)	320	320	97.330	97.815	41000.5	1.29544e-05	yes	nr
Video	Mean delay (ms)	320	320	21.805	12.045	70906.0	3.58992e-17	yes	nr
Video	Throughput (Mbps)	320	320	9.313	9.562	51556.5	0.879016	no	nr
Video	P99 delay (ms)	320	320	117.500	89.500	61614.0	8.4822e-06	yes	nr

4.2 Control flow by STA count

STA	KPI	p	Sig?	Better median	Mean LTE	Mean NR
5	PDR (%)	5.84998e-12	yes	lte	89.145	70.760
5	Mean delay (ms)	1.97678e-05	yes	nr	28.158	16.927
5	Throughput (Mbps)	5.37203e-12	yes	lte	0.614	0.487
5	P99 delay (ms)	0.0227257	yes	nr	230.975	150.275
10	PDR (%)	8.47292e-18	yes	lte	87.354	69.237
10	Mean delay (ms)	4.15396e-16	yes	nr	36.860	16.716
10	Throughput (Mbps)	8.84685e-18	yes	lte	1.179	0.934
10	P99 delay (ms)	1.63374e-12	yes	nr	384.925	166.750
15	PDR (%)	6.92753e-20	yes	lte	82.394	63.480
15	Mean delay (ms)	7.18055e-21	yes	nr	50.508	18.650
15	Throughput (Mbps)	7.14534e-20	yes	lte	1.633	1.258
15	P99 delay (ms)	1.71652e-19	yes	nr	528.200	184.675
20	PDR (%)	1.09558e-22	yes	lte	83.668	68.327
20	Mean delay (ms)	4.02558e-27	yes	nr	56.151	17.084
20	Throughput (Mbps)	1.13251e-22	yes	lte	2.164	1.767
20	P99 delay (ms)	7.27736e-26	yes	nr	665.962	172.363

5. Kruskal-Wallis — STA-count trend

Scope	Flow	Mode	KPI	H	p	Sig?	ρ (STA vs mean)	Means 5/10/15/20
by_flow	Control	ALL	PDR (%)	36.18	6.87051e-08	yes	-0.800	79.95/78.30/72.94/76.00
by_flow	Control	ALL	Mean delay (ms)	43.52	1.90475e-09	yes	1.000	22.54/26.79/34.58/36.62

Scope	Flow	Mode	KPI	H	p	Sig?	^P (STA vs mean)	Means 5/10/15/20
by_flow	Control	ALL	Throughput (Mbps)	548.66	1.35513e-118	yes	1.000	0.55/1.06/1.45/1.97
by_flow	Control	ALL	P99 delay (ms)	69.88	4.52591e-15	yes	1.000	190.62/275.84/356.44/419.16
by_flow	Sensor	ALL	PDR (%)	156.64	9.70967e-34	yes	-1.000	97.37/96.76/96.17/95.62
by_flow	Sensor	ALL	Mean delay (ms)	44.32	1.29199e-09	yes	1.000	23.19/25.92/30.77/31.97
by_flow	Sensor	ALL	Throughput (Mbps)	83.22	6.26214e-18	yes	1.000	5.30/7.61/8.93/9.93
by_flow	Sensor	ALL	P99 delay (ms)	51.31	4.1946e-11	yes	1.000	127.67/168.82/194.13/199.31
by_flow	Video	ALL	PDR (%)	93.05	4.85047e-20	yes	-1.000	98.04/97.69/97.34/97.01
by_flow	Video	ALL	Mean delay (ms)	33.75	2.24038e-07	yes	1.000	18.83/20.00/22.10/23.19
by_flow	Video	ALL	Throughput (Mbps)	193.56	1.04146e-41	yes	1.000	6.46/10.13/12.58/14.19
by_flow	Video	ALL	P99 delay (ms)	20.37	0.00014231	yes	1.000	108.66/128.93/138.25/140.72
by_flow_mode_lte	Control	lte	PDR (%)	47.64	2.54207e-10	yes	-0.800	89.14/87.35/82.39/83.67
by_flow_mode_lte	Control	lte	Mean delay (ms)	91.39	1.10421e-19	yes	1.000	28.16/36.86/50.51/56.15
by_flow_mode_lte	Control	lte	Throughput (Mbps)	295.68	8.55781e-64	yes	1.000	0.61/1.18/1.63/2.16
by_flow_mode_lte	Control	lte	P99 delay (ms)	123.53	1.3411e-26	yes	1.000	230.97/384.93/528.20/665.96
by_flow_mode_lte	Sensor	lte	PDR (%)	128.58	1.09215e-27	yes	-1.000	97.37/96.60/95.76/95.18
by_flow_mode_lte	Sensor	lte	Mean delay (ms)	56.11	3.98462e-12	yes	1.000	23.51/29.84/34.86/39.00
by_flow_mode_lte	Sensor	lte	Throughput (Mbps)	25.09	1.47993e-05	yes	1.000	5.73/7.65/8.52/8.95
by_flow_mode_lte	Sensor	lte	P99 delay (ms)	37.54	3.53491e-08	yes	1.000	129.21/183.18/207.05/213.01
by_flow_mode_lte	Video	lte	PDR (%)	72.63	1.16792e-15	yes	-1.000	98.04/97.54/97.08/96.71
by_flow_mode_lte	Video	lte	Mean delay (ms)	50.70	5.66281e-11	yes	1.000	19.26/22.67/25.70/29.24
by_flow_mode_lte	Video	lte	Throughput (Mbps)	73.59	7.2621e-16	yes	1.000	6.99/10.39/12.25/12.95
by_flow_mode_lte	Video	lte	P99 delay (ms)	20.04	0.000166176	yes	1.000	110.09/142.47/146.88/153.06
by_flow_mode_nr	Control	nr	PDR (%)	15.46	0.00146005	yes	-0.800	70.76/69.24/63.48/68.33
by_flow_mode_nr	Control	nr	Mean delay (ms)	3.46	0.326328	no	0.600	16.93/16.72/18.65/17.08
by_flow_mode_nr	Control	nr	Throughput (Mbps)	280.66	1.52399e-60	yes	1.000	0.49/0.93/1.26/1.77
by_flow_mode_nr	Control	nr	P99 delay (ms)	4.18	0.242982	no	0.800	150.28/166.75/184.68/172.36
by_flow_mode_nr	Sensor	nr	PDR (%)	45.49	7.26213e-10	yes	-1.000	97.37/96.92/96.59/96.06
by_flow_mode_nr	Sensor	nr	Mean delay (ms)	9.36	0.0248707	yes	0.600	22.88/22.00/26.68/24.95
by_flow_mode_nr	Sensor	nr	Throughput (Mbps)	57.83	1.70882e-12	yes	1.000	4.86/7.57/9.34/10.91
by_flow_mode_nr	Sensor	nr	P99 delay (ms)	16.97	0.000717116	yes	1.000	126.12/154.47/181.21/185.60
by_flow_mode_nr	Video	nr	PDR (%)	28.65	2.64897e-06	yes	-1.000	98.03/97.83/97.59/97.31

Scope	Flow	Mode	KPI	H	p	Sig?	^p (STA vs mean)	Means 5/10/15/20
by_flow_mode_nr	Video	nr	Mean delay (ms)	8.38	0.0388574	yes	-0.400	18.39/17.32/18.51/17.14
by_flow_mode_nr	Video	nr	Throughput (Mbps)	118.85	1.36711e-25	yes	1.000	5.93/9.88/12.90/15.42
by_flow_mode_nr	Video	nr	P99 delay (ms)	4.63	0.200759	no	0.800	107.22/115.39/129.62/128.38

6. Interpretation notes

- Bootstrap CIs quantify uncertainty after expanding from 3 → 10 seeds (640 matrix cells). The plan target is CI half-width $\leq 10\%$ of the mean for key KPIs.
- Mann-Whitney does **not** assume normality; it is appropriate for PDR and switching latency distributions.
- Kruskal-Wallis tests whether the four STA levels share one distribution; a significant result supports a STA-load effect without requiring linearity.

CSV artifacts: bootstrap_ci.csv, mann_whitney.csv, kruskal_wallis.csv, switch_bootstrap_ci.csv.

End of stats analysis — Sheikh Sayed Bin Rahman, PIC Lab, KIT